IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

Midterm Review

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Industrial & Management Engineering

Outline: Midterm Review

01 **Proposal Review**

02 **Midterm Review**

01 · Proposal Review

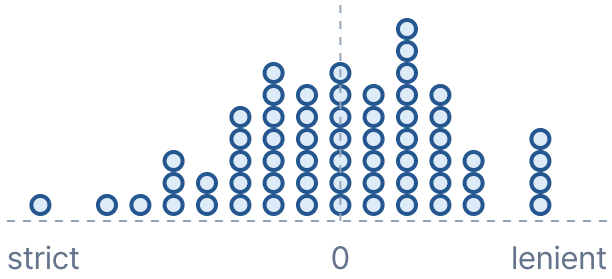
01 Proposal Review

02 Midterm Review

How Scores Were Processed

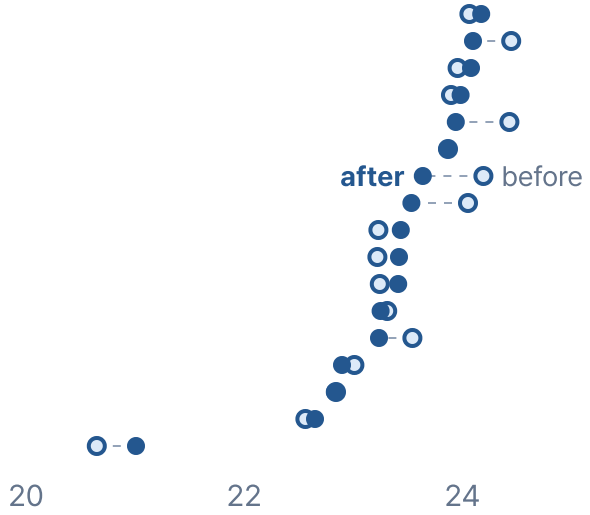
1. Measure rater strictness

per criterion: score = team + rater + error



61 raters, points on the 30-point total

2. Remove it from the scores

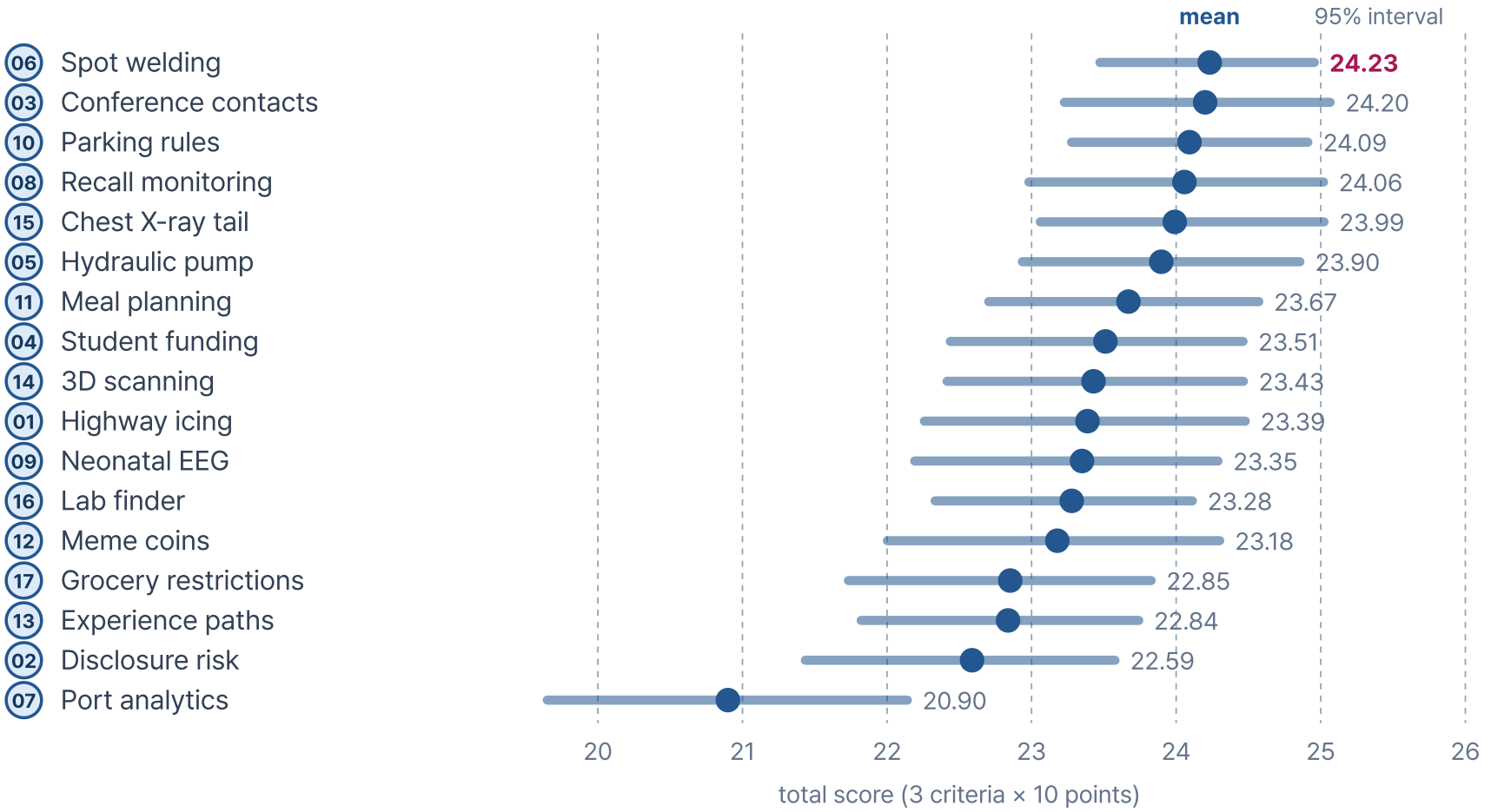


3. Weight the criteria

Problem	28%	×1.10
Method	17%	×1.33
Presentation	54%	×0.57

bar: share of the gap between teams
×: weight, one third each after

Peer Scores


mean
95% interval
06 Spot welding
03 Conference contacts
10 Parking rules
08 Recall monitoring
15 Chest X-ray tail
05 Hydraulic pump
11 Meal planning
04 Student funding
14 3D scanning
01 Highway icing
09 Neonatal EEG
16 Lab finder
12 Meme coins
17 Grocery restrictions
13 Experience paths
02 Disclosure risk
07 Port analytics
24.23
24.20
24.09
24.06
23.99
23.90
23.67
23.51
23.43
23.39
23.35
23.28
23.18
22.85
22.84
22.59
20.90
20
21
22
23
24
25
26
total score (3 criteria × 10 points)

Who Rated Well

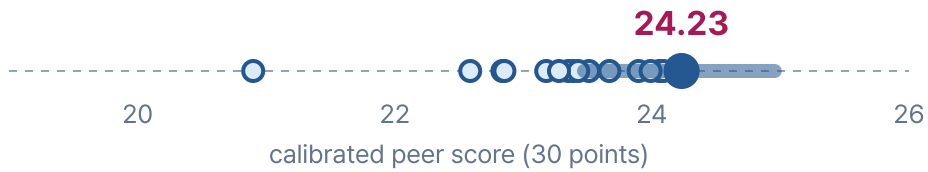
The 1st and the 2nd–3rd raters get bonus points



Team 06: Spot welding

Unsupervised anomaly detection on spot-welding logs





AI feedback on the slides, for reference only

WHAT WORKED

· Seven data issues diagnosed with numbers; 11,939 rows cut to 2,184 clean rows
· Working-day split proposed, with the 426-row anomaly as reference

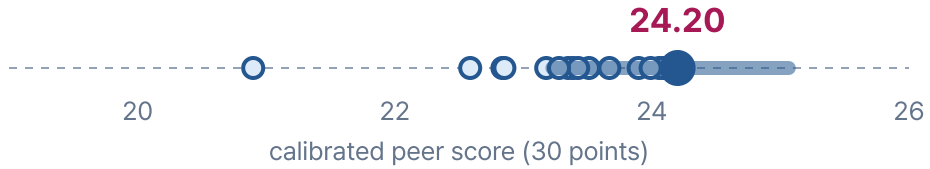
WHAT WAS MISSING

· Modeling columns defined, but no database engine or store chosen
· Modeling uses 2,184 rows without weld-level defect labels

Team 03: Conference contacts

Session and researcher matching from INFORMS programs and ORCID





AI feedback on the slides, for reference only

WHAT WORKED

· PostgreSQL design: 16 entities with keys, provenance and sync metadata
· Two concrete pipelines from INFORMS program HTML and ORCID API, with snapshot and upsert

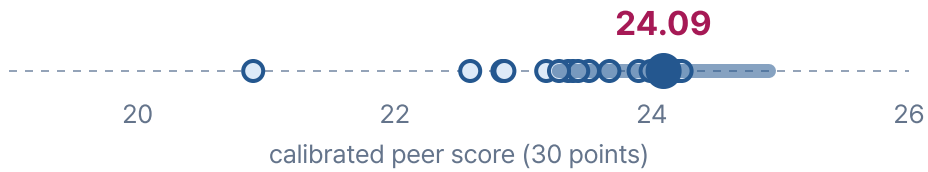
WHAT WAS MISSING

· Ingestion examples only; no loaded counts or executed query results
· No success metric or baseline for identity matching and session or researcher ranking

Team 10: Parking rules

Required vs actual parking from Pohang ordinances and building records





AI feedback on the slides, for reference only

WHAT WORKED

· Six public datasets joined to one row per building, 86,987 buildings, all counted
· Cleaning shown before and after in numbers, plus a first SQL result

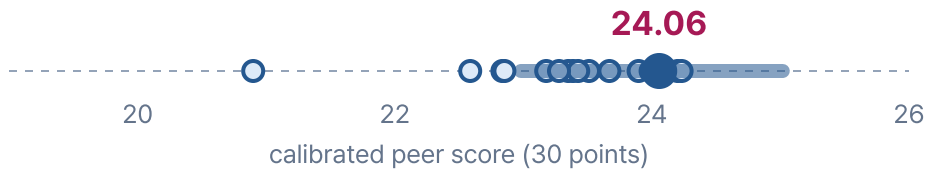
WHAT WAS MISSING

· Joined rows shown, but no database engine or primary keys
· 78% of parking records are 0 or missing; only 2,578 buildings compared so far

Team 08: Recall monitoring

Overseas recalls (44,518 records) matched to products sold in Korea





AI feedback on the slides, for reference only

WHAT WORKED

· 44,518 recall records from three APIs loaded into SQLite with four tables
· Image matcher piloted: 80% top-10 hits on 2,886 recall-to-recall queries

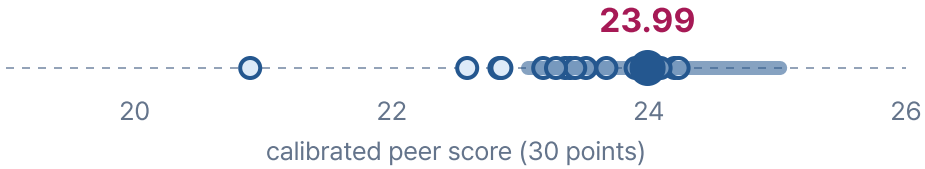
WHAT WAS MISSING

· Seven problem-framing pages before the built pipeline and results
· Korean listing side not collected: 30 hand-checked listings, no keys declared yet

Team 15: Chest X-ray tail

Long-tail analysis of 88,637 chest X-rays, with synthetic data for rare classes





AI feedback on the slides, for reference only

WHAT WORKED

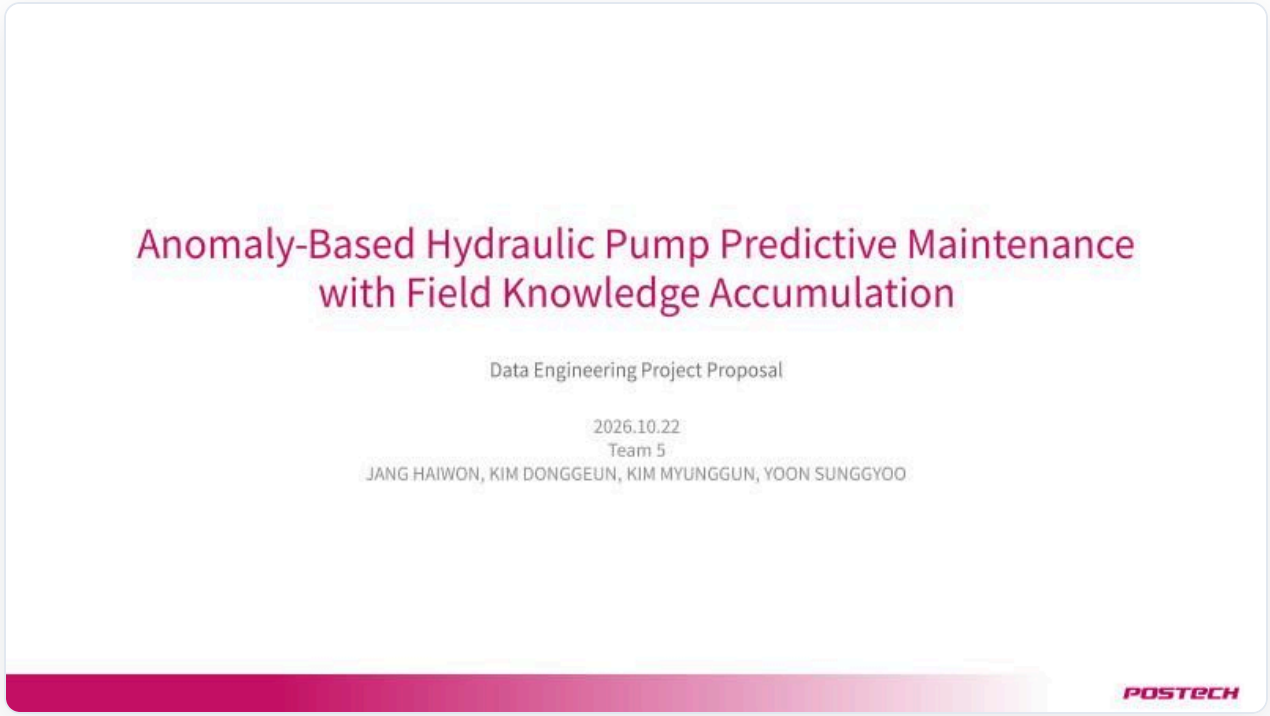
· Dataset profiled with numbers: 68,058 train images, 7 head classes hold 93.4%
· Baseline trained and measured, tail accuracy 0.000; schema with a real query

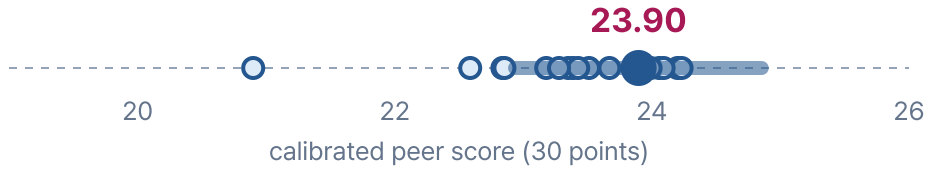
WHAT WAS MISSING

· Synthetic generation method unnamed; no check that generated images are valid
· Plan reports tail accuracy only; head-class trade-off unspecified

Team 05: Hydraulic pump

Pump anomaly alarms linked to maintenance memos through a knowledge graph





AI feedback on the slides, for reference only

WHAT WORKED

· Two-store design with candidates, selection criteria and four save checks
· Memo-link test cases defined: duplicate, revision, unknown cause, with ground-truth links

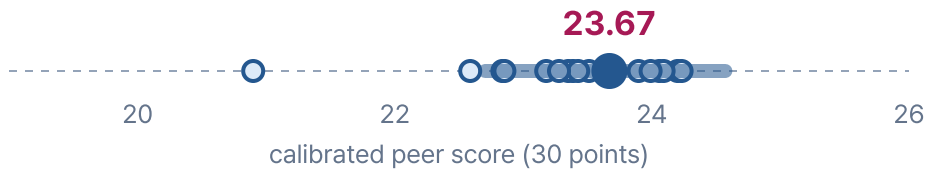
WHAT WAS MISSING

· No loaded sensor counts or results yet; demo still in preparation
· Memos are synthetic, written by the team; the knowledge side has no real data

Team 11: Meal planning

Meals and substitute recipes that respect dietary restrictions





AI feedback on the slides, for reference only

WHAT WORKED

· Four stores designed: recipe, ingredient, rules in SQL; user profile as document
· Clear user and pain: elimination-diet patients checking every meal for triggers

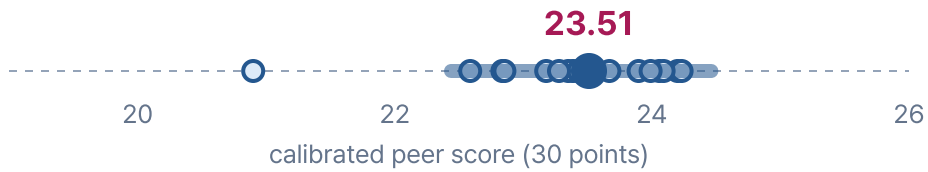
WHAT WAS MISSING

· Sample records only; collection volume and coverage not reported
· No success metric or baseline; five slides repeat one abstract flowchart

Team 04: Student funding

Funding notices matched to students, with drafts for the application





AI feedback on the slides, for reference only

WHAT WORKED

· One real notice already parsed into the JSONB schema with stored values
· Clear two-step deliverable: matched notices plus a first-draft outline

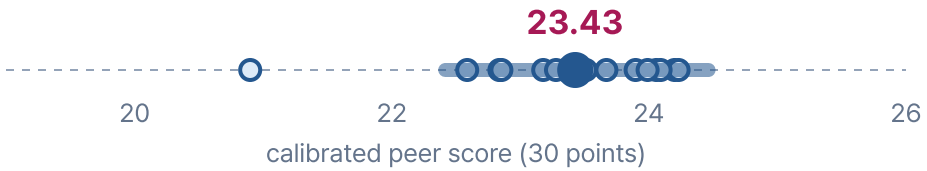
WHAT WAS MISSING

· Data so far is one program, three cases, five documents; no crawl count
· No success metric: how matching accuracy or outline quality is judged

Team 14: 3D scanning

Difficulty database of 3D shapes for curriculum-trained next-view scanning





AI feedback on the slides, for reference only

WHAT WORKED

· Real EDA already run: coverage versus views for bottle and dragon
· Difficulty DB with listed fields and a concrete SQL sampling query

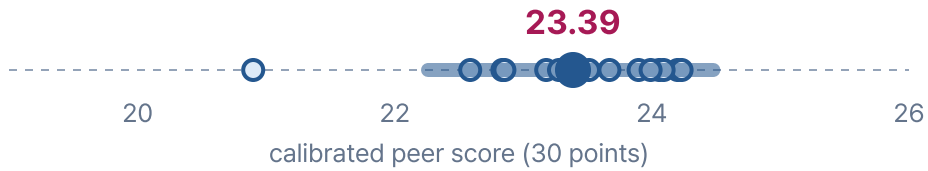
WHAT WAS MISSING

· Object dataset name and total size not specified
· Baselines unnamed, and slides 10-11 are dense diagrams hard to read

Team 01: Highway icing

Icing risk on unsensed highway stretches from road, weather and terrain data





AI feedback on the slides, for reference only

WHAT WORKED

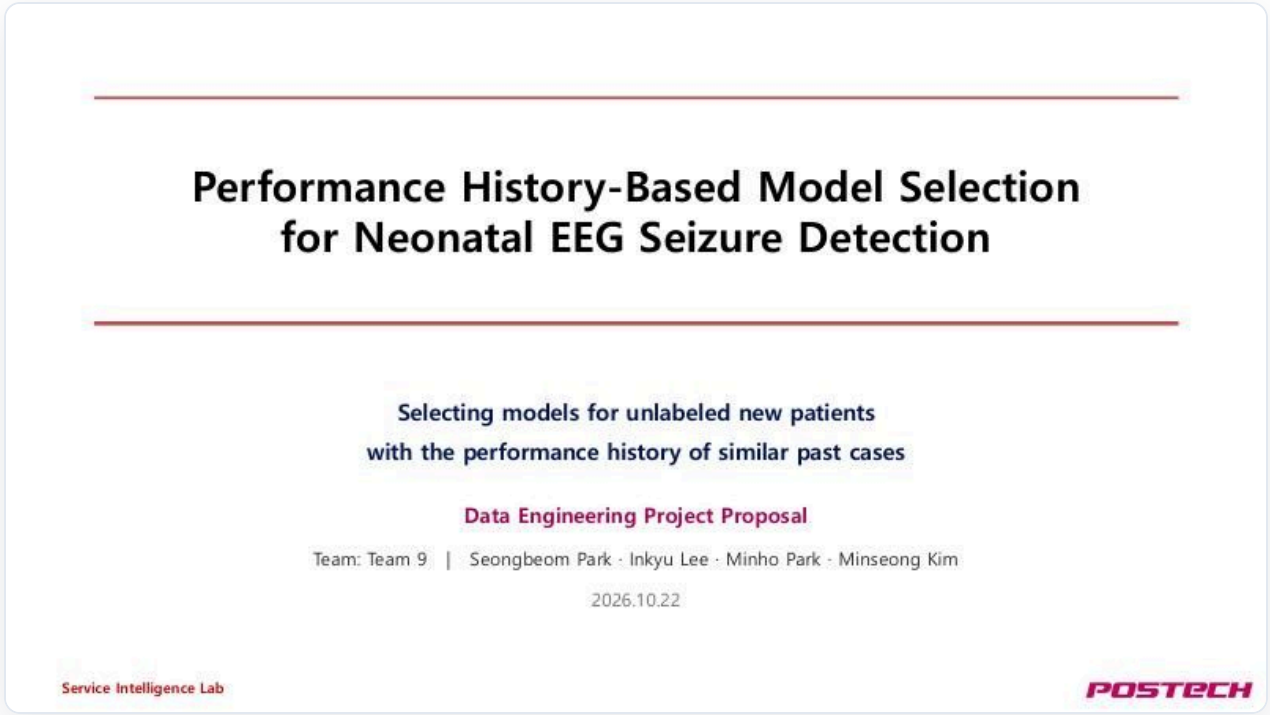
· Seven sources with cadence, three held-out sensors, baseline and leakage-safe CV
· Full PostGIS schema with keys, plus a shadow mask computed from real DEM

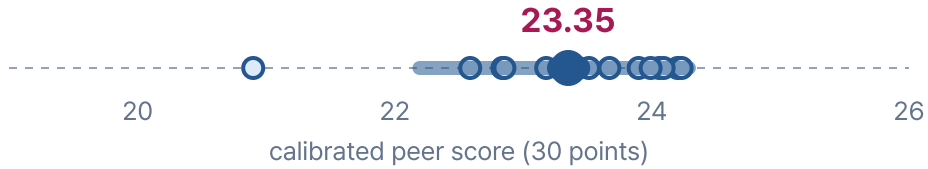
WHAT WAS MISSING

· Wide scope: terrain model, risk prediction, live map and LLM agents
· Loaded row counts and date range not reported

Team 09: Neonatal EEG

Seizure-model choice for a new newborn from similar past EEG segments





AI feedback on the slides, for reference only

WHAT WORKED

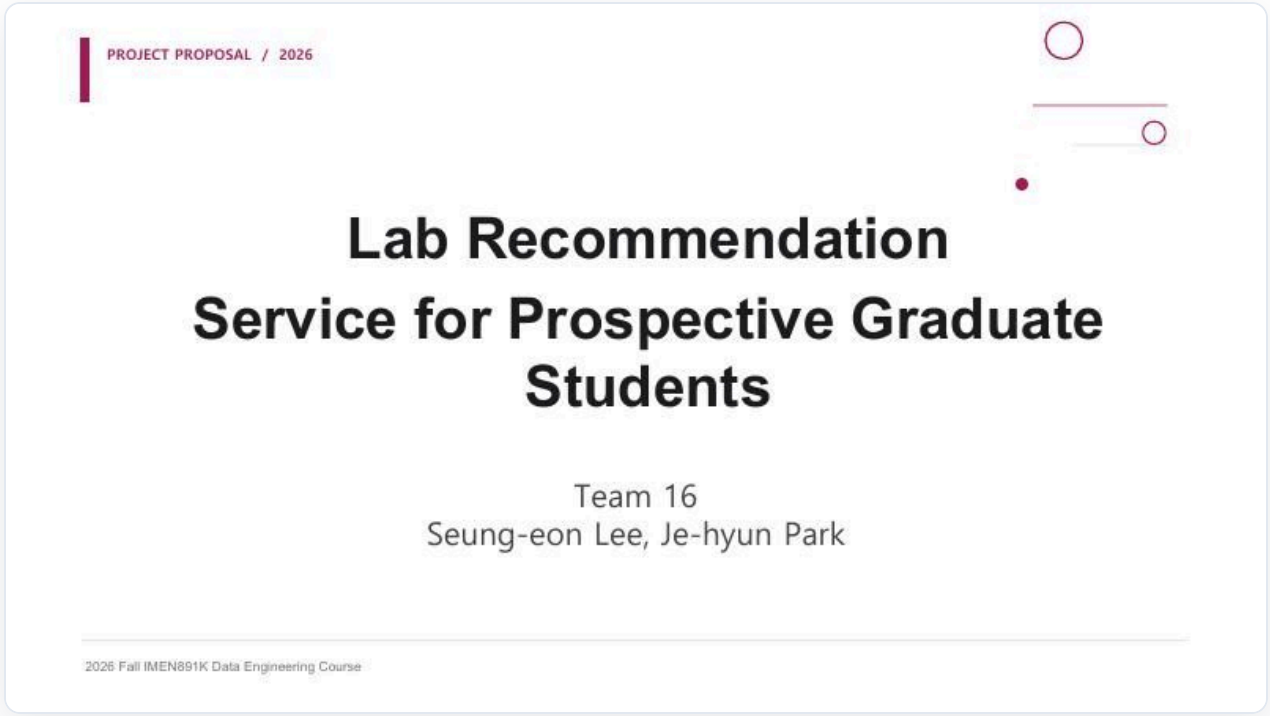
· Concrete public dataset: 79 neonates, 19 channels, 256 Hz, three annotators
· Patient-level 40/40/20 split, named baselines, PR-AUC and F1 metrics

WHAT WAS MISSING

· No model trained or selection performance measured yet
· Text-heavy slides with words broken mid-line hurt readability

Team 16: Lab finder

Lab recommendations from lab websites, papers and alumni





AI feedback on the slides, for reference only

WHAT WORKED

· Clear schema and raw-to-clean normalization rules with real DI page examples
· Polished UI mockup shows exactly what a student receives

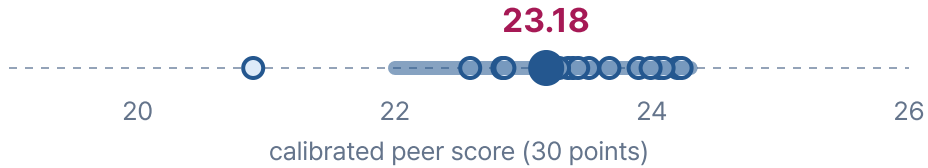
WHAT WAS MISSING

· Seven sample rows shown; full crawl coverage and totals unreported
· Fit score 95/100 undefined; no evaluation of recommendation quality

Team 12: Meme coins

Next-day direction for 10 meme coins from prices and social posts





AI feedback on the slides, for reference only

WHAT WORKED

· 14,773 daily price rows for 10 coins already crawled from CoinMarketCap
· Leakage-safe design: inputs frozen at 03:00 UTC before each forecast

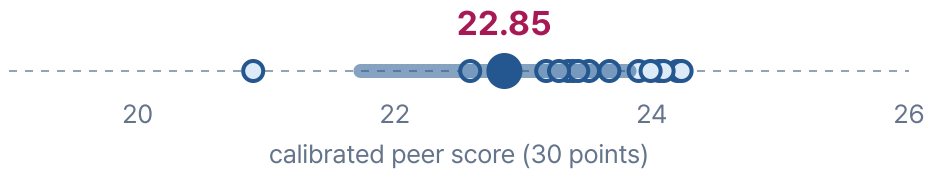
WHAT WAS MISSING

· Social data not collected yet; Reddit access still pending
· No forecast accuracy measured; dashboard content is illustrative

Team 17: Grocery restrictions

Products that fit each diet, from 6.2 M products across seven sources





AI feedback on the slides, for reference only

WHAT WORKED

· 6.2 million products collected from seven official APIs and open datasets
· SQL tables with row meaning; eligibility query measured 90 s to 1.8 s

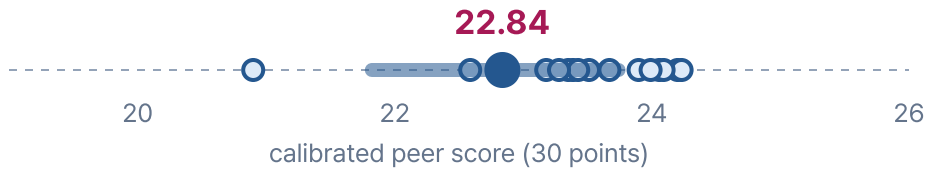
WHAT WAS MISSING

· Dense slides and wide scope: meal plans, chat, OCR on top of catalog
· Label accuracy still unmeasured; validation deferred to November

Team 13: Experience paths

Advice and papers for the next experiment from stored experience paths





AI feedback on the slides, for reference only

WHAT WORKED

· Clear worked example: filter-delay case shows what the system returns
· Column-level schema for experience steps and paper links

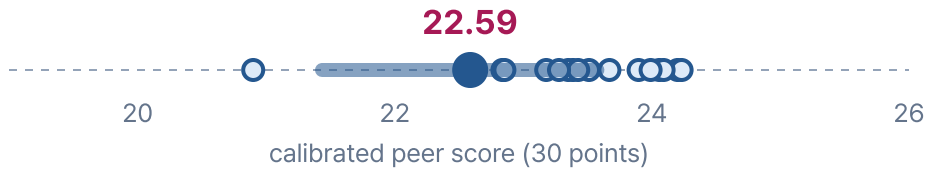
WHAT WAS MISSING

· No papers fetched or experience records counted yet
· Too abstract: four scenario pages, no executed retrieval result

Team 02: Disclosure risk

Risk scores for listed firms from OpenDART filings and KRX data





AI feedback on the slides, for reference only

WHAT WORKED

· Real OpenDART filings counted: 1,512 rights issues in one year
· Concrete cleaning steps shown: number casting, date formats, LLM extraction

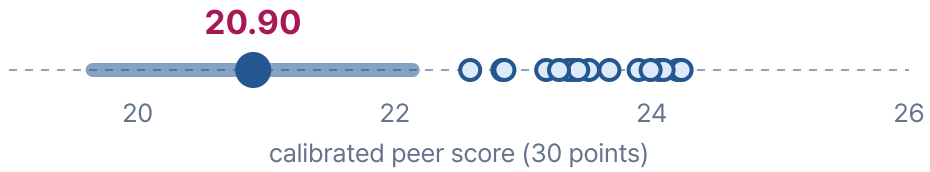
WHAT WAS MISSING

· Stores named, but no table definitions or keys shown
· Risk score formula undefined; dashboard is a fictional mockup; roadmap slide repeated

Team 07: Port analytics

Natural-language questions on Busan port statistics, answered by LLM agents





AI feedback on the slides, for reference only

WHAT WORKED

· Clear pain point: BPA portal serves fixed tables needing domain knowledge
· Terminology dictionaries gathered from MOF and MOLIT government files

WHAT WAS MISSING

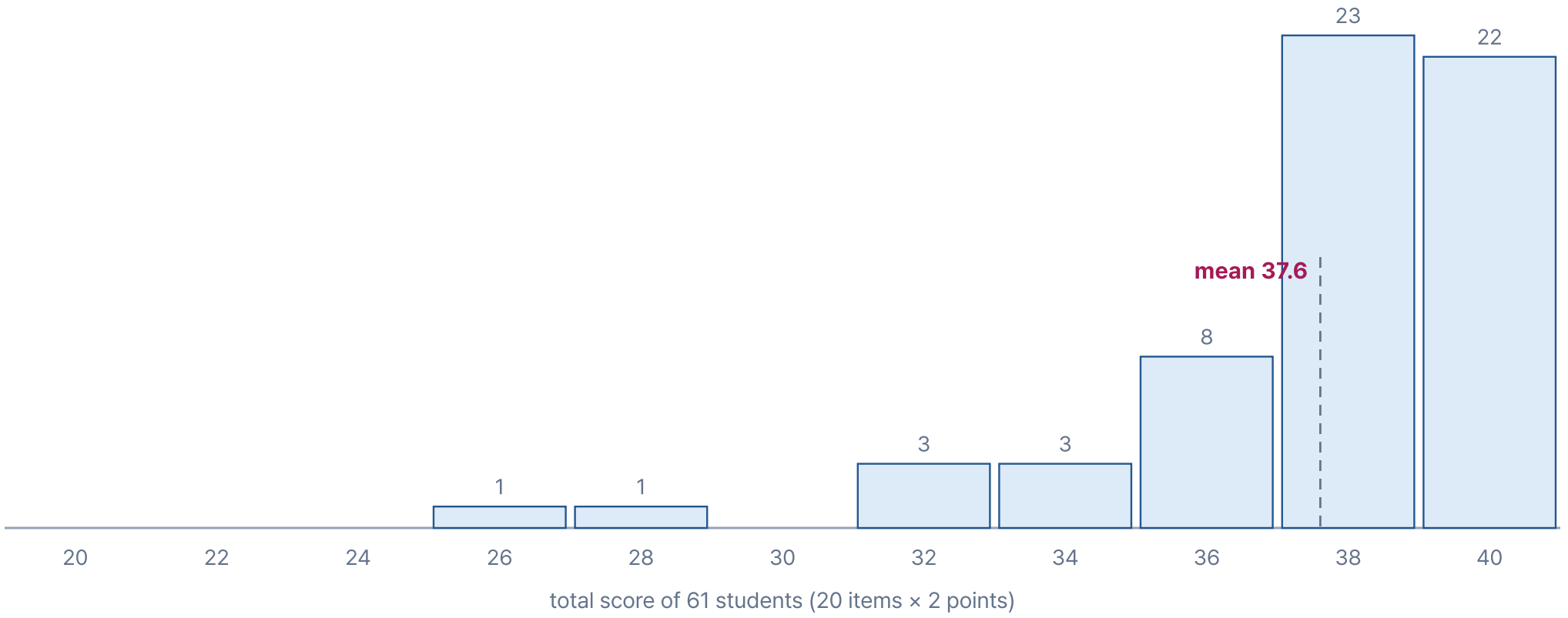
· Too abstract: agent workflow only, no port data loaded or schema
· No executed port query or measured result yet

02 · Midterm Review

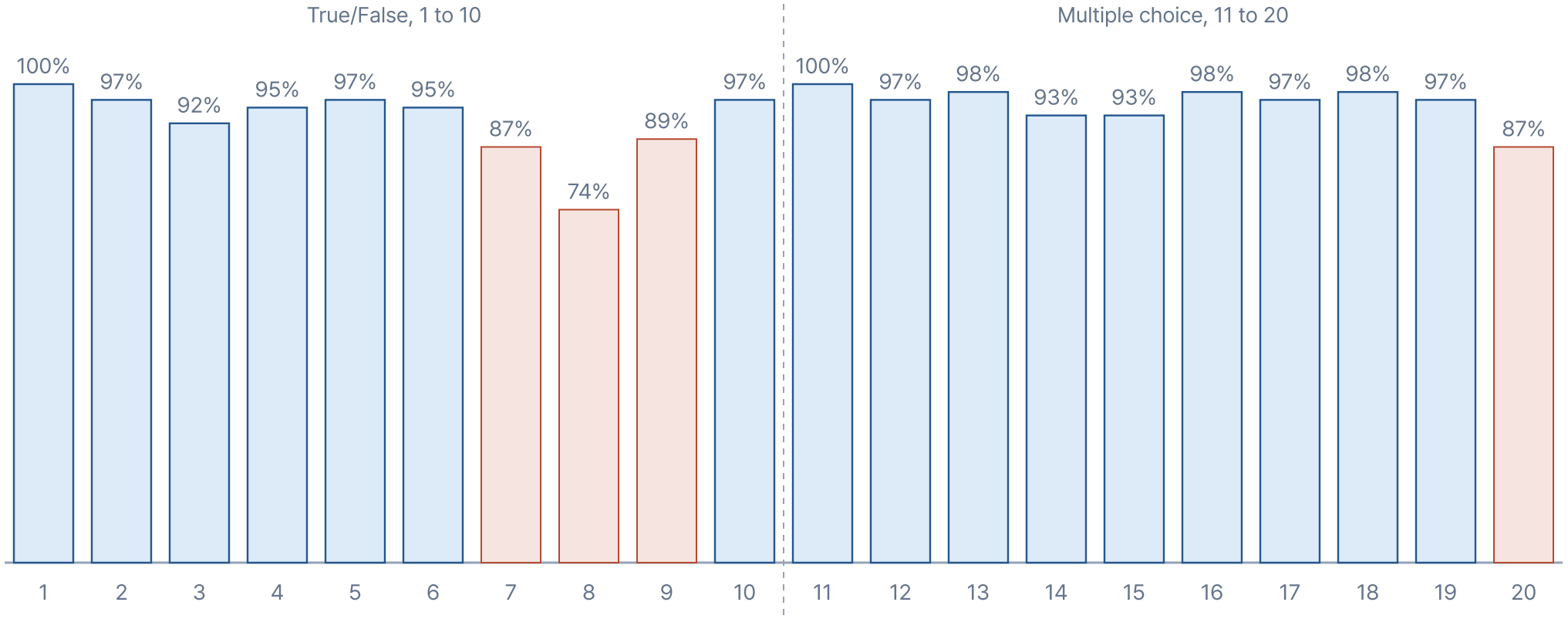
01 Proposal Review

02 **Midterm Review**

Score Distribution



Item Scores


True/False, 1 to 10
Multiple choice, 11 to 20
100%
97%
92%
95%
97%
95%
87%
74%
89%
97%
100%
97%
98%
93%
93%
98%
97%
98%
97%
87%
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19
20

Q1 Sensor as Encoder

A temperature sensor acts as an encoder: it converts a physical signal into a form that can be recorded digitally.

True ✓

False

100% right

Why. An instrument turns an event in the world into a record; a sensor encodes temperature as numbers

Q2 Correlation and Cause

Production volume and defect rate are strongly correlated. This proves that lowering production will lower the defect rate.

True

False ✓



97% right

Why. Correlation cannot say what happens under an intervention; a shared cause, such as rushed overtime, can drive both

Q3 Missing Values Before Filling

Some customer records have missing values. The team should fill them all before checking why they are missing.

True

False ✓



92% right

Why. Why a value is missing decides how to treat it; missingness itself can carry information

Q4 Schema Alignment vs Entity Matching

Two systems use different customer IDs. Giving their columns the same names is enough to match each customer.

- True
- False ✓



95% right

Why. Same column names align the schema; deciding which records are the same customer is entity matching, a separate step

Q5 Scaling Before Distance

A model measures customer similarity using age and income. Scaling helps stop income’s larger numbers from dominating the distance.

True ✓

False



97% right

Why. Distance adds squared differences; income in millions swamps age in tens. Scaling puts both on one footing

Q6 One-Hot and Order

A model must distinguish low, medium and high priority. One-hot encoding tells the model which level is higher.

True

False ✓



95% right

Why. One-hot gives each level its own column with no order between them; ordered levels need ordinal encoding

Q7 Selection vs Extraction

A team wants fewer inputs while keeping each variable's meaning. Feature selection does this by combining variables into new ones.

True

False ✓



87% right

Selection: keep a subset



Extraction: combine into new ones



Why. Selection keeps a subset of the original variables, so each keeps its meaning; combining them is extraction, such as PCA

Common slip: both reduce inputs, only extraction makes **new** ones

Q8 Keys Between Tables

A relational database can store each customer’s details once. Orders can refer to that customer using a key.

True ✓

False



74% right

Customer

id	name
c1	Kim
c2	Lee

Order

order	cust_id
o1	c1
o2	c1
o3	c2

Why. Customers live once in their own table; each order carries the customer key, a foreign key

Common slip: copying customer details into **every** order

Q9 Flexible Documents

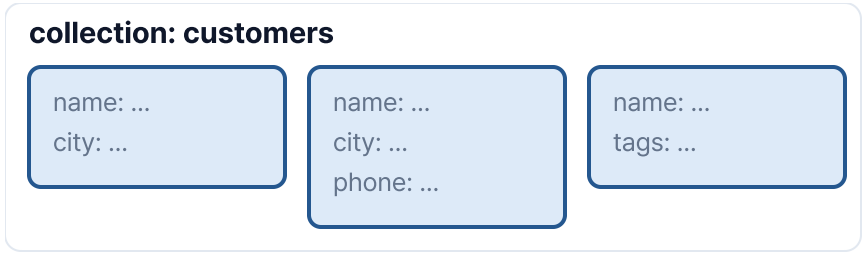
Some customer profiles have fields that others do not. A document store can keep these profiles in the same collection.

True ✓

False



89% right



Why. A collection does not force one schema; documents with extra fields sit side by side

Common slip: treating a collection like a table with **fixed** columns

Q10 Results as Inputs

A SQL query’s result can be a table in another query. A MongoDB aggregation stage can pass its documents to the next stage.

True ✓

False



97% right

Why. Both compose: a SQL result is a table another query reads; each stage hands documents to the next

Q11 API or Crawling

You need data from two sites: one has an API; the other only has web pages. Which methods fit the first and second sites?

A Web crawling; API calls

B API calls; web crawling ✓

C API calls; sensor measurement

D Synthetic data; web crawling

100% right

Why. Use the route the source offers: an API returns structured records; pages without an API need crawling

Q12 Typical Value Under Outliers

A few houses have prices far above the rest. Which measure of a typical price is less affected by these extreme values?

A Mean

B Range

C Standard deviation

D Median ✓



97% right

Why. The median depends on order, not size: a few huge prices move the mean, not the middle value

Q13 Same Statistics, Different Shapes

Two stores have the same means, variances and correlation for sales and ad spending. Why still compare their scatter plots?

A Matching statistics mean the data have the same shape

B Matching statistics mean the data have the same outliers

C Different patterns can produce the same statistics ✓

D A scatter plot cannot show anything beyond the correlation



98% right

Why. Anscombe's quartet: four datasets share mean, variance and correlation yet look completely different

Q14 Smoothing a Noisy Trend

Sensor readings fluctuate rapidly because of noise. You want a smoother trend on a dashboard. Which method fits?

A Moving average ✓

B One-hot encoding

C kNN imputation

D Z-score standardization



93% right

Why. A moving average replaces each point with the mean of its neighbors, damping short-term noise

Q15 Sampling Every Factory

You need a sample that includes every factory. Some factories have far more records than others. What should you do?

A Take the first records in the combined dataset

B Sample only from the largest factory

C Randomly sample at least one record from every factory ✓

D Remove unusually large or small values



93% right

Why. Group by factory and draw at least one record from each, so no factory is left out

Q16 SELECT-FROM-WHERE

This query builds a product list. What does the list contain?

```
SELECT name
FROM Product
WHERE category = 'Electronics';
```

A Names of Electronics products ✓

B All fields of Electronics products

C Names of all products

D The number of Electronics products



98% right

Why. SELECT picks the name column; WHERE keeps only Electronics rows

Q17 Basic JOIN

Every product has a matching company. This query builds a supplier report. What does it show?

```
SELECT p.name, c.country
FROM Product p JOIN Company c
  ON p.company_id = c.company_id;
```

A The number of products in each country

B Every possible product–company pair

C Each product and its category

D Each product and the country of its company ✓



97% right

Why. ON pairs each product with the company whose ID matches; company_id is unique, so one row per product

Q18 GROUP BY and Average

A dashboard runs this query. What does each result row describe?

```
SELECT category, AVG(price) AS average_price
FROM Product
GROUP BY category;
```

A Each product and its price

B One average price for all products

C Each category and its average product price ✓

D Each company and its average product price



98% right

Why. GROUP BY category makes one row per category; AVG(price) is computed inside each group

Q19 Access Pattern to NoSQL Type

A service must (i) find a session by its token and (ii) follow links between friends. Which types fit, in that order?

A Graph; key-value

B Key-value; graph ✓

C Document; wide-column

D Wide-column; document



97% right

Why. A token lookup is a get by key: key-value. Following friend links is traversal: graph

Q20 Aggregation Stages

Each sale records a city, product category and amount. You need total Electronics sales in each city. Which steps?

A Keep Electronics; group by category; sum amounts

B Keep Electronics; group by city; count the sales

C Group all sales by city; sum amounts; sort totals

D Keep Electronics; group by city; sum amounts ✓



87% right

$match category = Electronics

$group _id: city, total: $sum amount

one row per city; $sum works inside $group

Why. Filter first ($match), then group by city ($group), then sum the amounts ($sum)

Common slip: grouping by **category** gives one total, not one per city

Thank you for your hard work on the proposal and midterm!